

Absorption Hold-Check: Requirements

Product/Business Requirements · Martin Ignatov · 25 Sep 2026

Product Requirements

1. Problem

Order-flow traders watch for absorption: a large resting order quietly absorbing aggressive selling or buying while price refuses to move through it, often just ahead of a pullback or reversal. Our partner teaches this pattern daily and still misses about half of it live.

Bookmap already ships an Absorption indicator (desktop, Global+), but it only measures volume: it keeps counting after price trades straight through a level, so it can't tell a level that held from one that broke. It needs per-instrument tuning with no ES/CL presets, and its alerts need self-built Telegram bot. Third-party add-ons exist too, at extra cost, and don't close these gaps.

The partner wants this fixed, tuned for ES and CL, and credited as her method, before her cohort starts in 6 weeks.

Competitors (ATAS, Quantower, DeepCharts) already ship similar order-flow indicators, so this is also about not falling behind on a feature our own paying users already expect.

2. Goals, metrics and value

Goal	How we will know
Catch what the partner teaches	Correctly catches $\geq 80\%$ of labelled real examples, reviewed with the partner (with a confidence range, not a single number)
Alerts worth listening to	$\geq 70\%$ alerts rated valid in review
Zero-setup start	A new user has working alerts in ≤ 3 clicks and ≤ 60 seconds, typing nothing
Ready in time	Temporary fix live in week 1; full build ready 5 working days before cohort day 1 (week 6)
Build once	One specification and one test suite serves web now and desktop later

Non-goals: predicting reversals or scoring probability, buy/sell signals, placing orders, win-rate or P&L statistics, replacing existing tools, or a partner-exclusive build.

Value, roughly: cohort size $\times$ trial-to-paid rate $\times$ (plan + data fees) $\times$ months retained, plus a retention lift for existing Absorption users, against about six weeks of a small engineering team. Kill or defer criterion: if past partner cohorts were under 50 students or converted under roughly 10%, ship only the temporary fix and put the full build on the normal roadmap instead.

Targets above get finalised once we have baselines: past cohort size and conversion from affiliate and billing data, today's Absorption usage and alert-enable rate from product telemetry, and demand signal from marketplace sales of similar add-ons.

3. Who it is for

- Partner educator — needs her own thresholds to fire live on stream while teaching, and credit for her method.
- Cohort student — needs it working with zero configuration, seeing exactly what the educator sees.
- Existing Bookmap trader — needs fewer false alarms on levels that break, and alerts that work without setup.

Stakeholders outside the immediate build: Partnerships (the relationship with the partner), Support (today's alert-defect tickets), the Market Pulse Web team (this feature will live inside their product), and Compliance (alert wording and disclaimers).

4. What we are building

Web, week 6 — the cohort-facing MVP

- Detection logic from the Introduction — roughly 600 ES / 300 CL contracts absorbed within 60 seconds, holding for 3s (ES) or 5s (CL) through a short confirm window, tolerating a brief 1-tick poke that price immediately reclaims, and using a wider 3-tick band for CL's thinner book — with built-in ES and CL presets.
- Each event tracked on the chart from Absorbing through Reacting to a final Held, Broken or Expired outcome, so traders learn from what actually happened.
- One Low/Medium/High sensitivity dial for most users, with an advanced panel underneath for power users, safe input limits, and a reset button.
- Alert hours per preset, so alerts only fire when they matter (e.g. the partner's own morning session).
- The partner's ES/CL presets bundled in and credited to her by name; settings saved to the user's account.
- Zero-setup alerts — pop-up, sound, browser notification — capped so a noisy stretch can't spam anyone.
- Today's existing Absorption indicator keeps working unchanged for everyone who doesn't opt in.
- One detector per instrument, shared by every viewer — a whole cohort watching ES costs the same as one person watching it.
- Enough usage tracking to know if it is working: who turns it on, who mutes it, and whether alerts get clicked.

Cut first if week 3 shows we are behind, in this order

A live "what would have fired" preview → thumbs-up/down feedback on alerts → a "building" visual before an alert fires → in-chart toast (falls back to existing alerts) → preset sharing via file or code → an auto-tuning threshold for other instruments → news-time muting.

Phase 2, after week 6

- The same logic ported into today's desktop Absorption indicator.
- Order-level (MBO) confidence scoring, reusing the existing Stops & Icebergs engine.
- Push notifications, one-click Telegram, and e-mail.
- A public, signed gallery for sharing presets beyond ES/CL, and tuning for instruments beyond ES/CL.

Out of scope

Predicting reversals or scores, auto-trading, win-rate or P&L statistics, a partner-exclusive build.

Temporary fix, week 1, no code

We manually configure today's existing desktop tool for the partner's own ES/CL trading, with sound and pop-up alerts, so she has something working immediately while the real build is underway. Exit criterion: an alert fires live on her own machine and connection.

5. How we will know it worked, before launch

The partner labels at least 5 real events per instrument — a reasonable ask given her time — while a Bookmap educator independently labels a larger sample, at least 25 per instrument, for statistical confidence. Both label blind, across both a calm day and a volatile one. We tune our defaults on most of that data and check accuracy on days we held back for testing. We only launch once the accuracy and alert-quality goals above are met and both the Head of Product and the partner sign off. Defaults stay provisional for the first two weeks of live cohort feedback before we lock them in as the confirmed defaults.

After launch we track whether people who turn it on get an alert, click it, and are still using it after 7 and 30 days, for the cohort versus everyone else. The mix of events that end Held versus Broken is watched only as a quality signal, never published as a win rate or success claim.

6. Key risks

Risk	Mitigation
Timeline slips — effort is hard to size before we're in the codebase	The temporary fix ships in week 1 regardless; a checkpoint in week 3 decides using the cut list above
The web platform can't host a new detector the way we assume, or ES/CL data isn't available there (our biggest week-0 unknowns)	Confirm both in week 0; fallback is a standalone settings screen, or an ES-only launch if CL data isn't ready in time
Our definition of "absorption" does not match the partner's	Independent blind labelling and a joint sign-off before launch
Alert fatigue, either too many or too few	The confirm window, a cap on alerts per minute, and alert-hours are built in from day one
Over-claiming what the feature can predict	Descriptive-only alert copy, a disclaimer on first use, and a partner review
Partner dependency and channel conflict with other educators	Keep the feature generic; credited presets are open to any verified educator, not exclusive to this partner
Plan-tier confusion — Absorption is Global+ only, but some students may be on a lower plan	Decide how we handle students without the right plan in week 1; show a clear "start trial" state instead of hiding the feature

7. Release plan

When	What happens
Week 0	Align scope with the Head of Product and engineering; confirm the web platform can support this; start the partner conversation
Week 1	Temporary fix live for the partner; the real detection build starts; blind labelling begins
Weeks 2–3	Detection engine complete and tested; markers, settings and alerts built; checkpoint to decide whether to continue as planned
Weeks 4–5	Partner beta; defaults calibrated from real labelled data; training materials ready
Week 6	Cohort launch — 5 working days before the cohort's day 1
Phase 2 (weeks 7–10)	Desktop parity, so existing Global+ users get the same upgrade

Rollout is staged — partner, then cohort, then a wider beta, then everyone — so a web release can be pulled back instantly if something looks wrong. The main dependencies are the partner's and an educator's time for labelling in weeks 1–2 (the step that can't slip), plus about a week of design time and the QA test fixtures.

8. Compliance and copy

- **Describe, don't direct:** "absorbing", "level held", "level broken". Never "buy", "sell", "reversal signal" or "high probability".
- **Disclaimer** on first enable and in the KB: "An order-flow analytics aid, not a trade signal or recommendation. Absorption doesn't guarantee a reversal. Trading futures involves substantial risk of loss."
- **Partner wording:** "«Partner»'s ES/CL settings for Bookmap's Absorption indicator". No performance claims; Bookmap reviews the partner's materials.
- **Badge wording:** "Published by a verified educator account". It verifies the publisher, not results.

Business Analysis

The rest of this document is the full technical specification: exact detection-engine rules and parameters, non-functional requirements, and the 28-scenario acceptance-test suite — enough for a development team to build and test against without coming back for interpretation. It is implemented and passing in the working prototype (03).

Reading the sections below: (M) = must-have for the week-6 MVP; (S) = should-have, cut first in the order listed in Product Requirements §4; (P2) = Phase 2. [D] = desktop-only, [W] = web-only; untagged applies to both. “Eng to confirm” flags an assumption about platform capabilities (see §5). Values marked “calibrate” are starting points the partner and an educator will tune from real labelled data (Product Requirements §5), not fixed constants.

1. Definitions

Term	Definition
Message	One input update from the feed (depth or trade). Distinct from an <i>event</i>
Feed time	The timestamp Bookmap attaches to each message and records in replay files. All detection logic uses feed time; wall-clock time is never used
Tick	The instrument's price step as delivered to the engine. Prices are stored as integer tick indices. ES 0.25 (\$12.50) · MES 0.25 (\$1.25) · CL 0.01 (\$10) · MCL 0.01 (\$1)
Side / aggressor	A bid level holds passive buyers; an ask level holds passive sellers. The aggressor is the side of the market order: SELL hits the bid, BUY lifts the ask
Band	The price range of a level: <code>zone_ticks</code> wide from its anchor (the first qualifying trade's price), extending the way aggressors push (down for bid, up for ask). The near edge is the anchor; the far edge is the other end. ES band = 1 tick
Qualifying trade	SELL at a price inside an open bid band, or BUY inside an open ask band
Episode	A run of qualifying trades in one band (FR-DET-05...07)
V_win / V_total	Qualifying volume in the rolling window / in the whole episode
Displayed size	Aggregated resting size at a price. S0 = at the anchor after all depth messages strictly earlier than the first trade. S_max = maximum over depth messages during the episode (trades don't change it)
Replenishment ratio	$V_total \div \max(S_max, 1)$: how many times the level absorbed its largest shown size. It is not a claim about hidden orders on MBP
Poke / reclaim	A trade 1...(break_ticks - 1) ticks beyond the far edge is a poke. It is reclaimed by any trade back inside the band (or on the absorbing side) within <code>reclaim_s</code>
Event	A detection record (FR-DET-09). States: ABSORBING and REACTING are live; HELD , BROKEN , EXPIRED and INTERRUPTED are final
Reasons	HELD: MOVED_AWAY. BROKEN: TRADED_THROUGH, NOT_RECLAIMED, PULLED (S). EXPIRED: TIMEOUT. INTERRUPTED: DATA_GAP, USER_STOP, FEED_CHANGE, SEEK, ENGINE_ERROR, SESSION_END
Session	The CME trading day, 18:00–17:00 ET
Other	SD: standard deviation (today's Auto threshold = session mean + SD × multiplier). RTH: regular trading hours. TAS: trade at settlement. MBP / MBO: market-by-price / market-by-order

2. Detection engine: Hold-Check mode (FR-DET)

Parameters (per preset; volumes in contracts; all values *calibrate*)

Parameter	Meaning	ES	CL	Allowed
min_volume	V_win needed to reach the threshold	600	300	1–100 000
window_s	Rolling window for V_win	60	60	5–600
max_gap_s	Episode ends after this long with no qualifying trade	20	20	1–120
zone_ticks	Band width	1	3	1–10
confirm_s	After the threshold, the level must survive this long before the event	3	5	0–30

min_ratio / min_remaining	Holding evidence: replenishment ratio $\geq$ min_ratio, or displayed size at the level $\geq$ min_remaining $\times$ S0	1.0 / 50%	1.0 / 50%	0–10 / 0–100%
break_ticks	A trade this far beyond the far edge breaks it at once	2	3	1–20
reclaim_s	A poke must be reclaimed within this time	3	3	1–30
min_visible	Optional: S_max must reach this ("big visible order"). 0 = off	0	0	0–100 000
react_ticks / held_ticks	Move away from the near edge for Reacting / Held (distance-based, not a timer: a fast move away still counts)	4 / 8	5 / 15	1–100 / 1–200
outcome_timeout_s	Time allowed for Held or Broken before Expired	300	300	10–3 600
sensitivity	Multiplies min_volume and min_visible: Low 1.5, Medium 1.0, High 0.6. Effective value = $\max(1, \text{ceil}(v \times m))$	Medium	Medium	
direction	Both / Buyers absorbing / Sellers absorbing	Both	Both	
alert_hours	Alerts fire only inside this window (ET)	09:30–16:00	09:00–14:30	

Processing and boundary conventions

- **FR-DET-01 (M)** There is one engine per instrument, hosted server-side (the Market Pulse Web algorithm-hosting model) and shared by every connected client watching that instrument, fed by that instrument's ordered messages. Desktop parity reuses the same engine inside the add-on framework (Phase 2).
- **FR-DET-02 (M) Deadlines.**
 - Before processing a message at feed time t , the engine fires every pending deadline **strictly earlier than t** , in deadline order, each stamped at its deadline.
 - Ties go in this order: event outcome deadlines (timeout, reclaim), then confirm deadlines, then episode deadlines (reclaim, gap), then lower price first.
 - A message **at exactly** a deadline is processed first. So a trade at exactly $t_{\text{last}} + \text{max_gap_s}$ continues the episode, and a trade at exactly the timeout can still produce Held or Broken.
 - In quiet markets, deadlines also fire from the platform's feed-time heartbeat (*Eng to confirm*), never from wall-clock time.
- **FR-DET-03 (M) Per trade, in order:**
 - (a) outcome checks for live events, using their snapshot parameters;
 - (b) break and poke checks for open episodes;
 - (c) accumulation;
 - (d) the threshold check.

Trades with unknown aggressor are never accumulated, but they count for (a) and (b). Trades the feed flags as non-regular (block, spread leg, TAS) are ignored (*Eng to confirm* the flag).

- **FR-DET-04 (M)** The window is $(t - \text{window_s}, t]$. A lot exactly window_s old is excluded.

Rules

- **FR-DET-05 (M) Episode start.** A qualifying trade outside every open band on its side opens an episode anchored at its price. It records t_{start} , S0 and $S_{\text{max}} = S0$. A trade inside several bands joins the oldest.
- **FR-DET-06 (M) Accumulation.** Each qualifying trade adds to V_{total} and the window. $t_{\text{last}} =$ its time.
- **FR-DET-07 (M) Episode end.** An episode ends at the earliest of:
 - (i) $t_{\text{last}} + \text{max_gap_s}$ (GAP);

- (ii) a trade $\geq \text{break_ticks}$ beyond the far edge (TRADE_THROUGH);
- (iii) a poke not reclaimed within reclaim_s (NOT_RECLAIMED, stamped at the deadline);
- (iv) interruption.

A reclaimed poke adds 1 to pokes and the episode continues. An episode that ends without an event is discarded: no marker, alert or log entry.

- **FR-DET-08 (M) Threshold.** On each qualifying trade in an episode with no event and no pending confirm, check all of these:
 1. $V_win \geq \text{effective min_volume}$;
 2. holding evidence ($V_total \div \max(S_max, 1) \geq \text{min_ratio}$, or current displayed size $\geq \text{min_remaining} \times S0$);
 3. min_visible is 0, or $S_max \geq \text{effective min_visible}$;
 4. the direction filter allows the side;
 5. **one live event per level:** there is no live event on the same side whose band is within 1 tick.

If all hold, record t_cross and V_cross and start the confirm window. A blocked episode is re-checked on every qualifying trade.

- **FR-DET-09 (M) Event.** At $t_cross + \text{confirm_s}$, if the episode is still open and no poke is pending, create the event: state ABSORBING, $t_trigger = \text{that deadline}$. If a poke is pending, create it at the reclaiming trade. The event records:
 - $\text{id} = \{\text{instrument}\}|\{\text{side}\}|\{\text{price_ticks}\}|\{t_trigger_ns\}$;
 - $V_trigger = V_cross, S0, S_max, \text{pokes}$;
 - test_n : previous event on an overlapping band reached Reacting or Held $\rightarrow$ its $\text{test_n} + 1$; otherwise 1;
 - a preset snapshot (id, version, author, every effective parameter).

This fires the Absorbing alert (FR-ALR).

- **FR-DET-10 (M) Outcomes**, using snapshot parameters:
 - A trade $\geq \text{break_ticks}$ beyond the far edge $\rightarrow$ **BROKEN** (TRADED_THROUGH).
 - A poke not reclaimed within $\text{reclaim_s} \rightarrow$ **BROKEN** (NOT_RECLAIMED), at the deadline.
 - A trade $\geq \text{react_ticks}$ beyond the near edge on the away side $\rightarrow$ **REACTING** (live).
 - A trade $\geq \text{held_ticks}$ away $\rightarrow$ **HELD**.
 - $t_trigger + \text{outcome_timeout_s} \rightarrow$ **EXPIRED** (TIMEOUT).
 - **S:** BROKEN reason is PULLED if, over $[t_break - 2\text{ s}, t_break)$, the maximum displayed size at the level minus the size just before t_break , minus the qualifying volume in that interval, is $\geq 0.5 \times S_max$.
- **FR-DET-11 (M) Freeze.** Once an event is final, V_total , S_max and t_last are frozen. $t_end =$ when the episode closed, or t_state if the event was final first. $\text{end_reason} \in \{\text{GAP}, \text{TRADE_THROUGH}, \text{NOT_RECLAIMED}, \text{DATA_GAP}, \text{FINALIZED}\}$. An episode left open only blocks new events at its level; it never updates the event.
- **FR-DET-12 (M) Other endings.** Disabling the indicator, switching mode or preset, unsubscribing the instrument, a Replay seek, an engine error or a contract change turns live events INTERRUPTED with the matching reason.
- **FR-DET-13 (M) Parameter changes** take effect at the next message. Open episodes without an event are discarded and restart. Live events keep their snapshot.
- **FR-DET-14 (M)** Derived fields for the tooltip, log and schema: duration ($t_last - t_start$), V_total , $V_trigger$, $S0$, S_max , replenishment ratio, pokes, test_n , time to outcome, and maximum move away in ticks. **S:** net delta at the band during the episode, and peak and current hits/s (the exhaustion cue).
- **FR-DET-15 (M)** A level flip (the ask at p consumed, then bids form at p) is not a break in v1.

How the rules handle the counter-patterns Bookmap teaches

Situation	Result
Threshold reached, then traded through within the confirm window (breakout)	No event, no alert (AC-03). Classic would mark it
1-tick stop-run poke that reclaims	The episode or event continues, poke logged (AC-04, AC-06)
Level eaten through (visible size falls, no refill)	No holding evidence, so no event (AC-11)
Pulled or spoofed wall (no trades)	No qualifying volume, so no event. Pulled after detection, then traded through $\rightarrow$ BROKEN (PULLED, S)
Sweep through several levels	Each level breaks. No event
Exhaustion (little volume)	Below threshold. No event
Iceberg with no visible size	Detected on volume plus refill; high replenishment ratio
CL absorption across a few ticks	Summed across the band (AC-23)

Retest after a reaction	New event labelled Test 2 (AC-14)
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Lifecycle

Object	From → to	Trigger
Episode	(none) → OPEN	First qualifying trade outside open bands
Episode	OPEN → confirming	Threshold conditions met (FR-DET-08)
Episode	confirming → event	Confirm window survived (FR-DET-09)
Episode	OPEN/confirming → discarded	GAP, TRADE_THROUGH, NOT_RECLAIMED or interruption before the event
Event	ABSORBING → REACTING	Price ≥ react_ticks away
Event	ABSORBING/REACTING → HELD	Price ≥ held_ticks away (final)
Event	ABSORBING/REACTING → BROKEN	Trade-through, or unreclaimed poke (final)
Event	ABSORBING/REACTING → EXPIRED	Timeout (final)
Event	ABSORBING/REACTING → INTERRUPTED	Data gap, user stop, seek, error, session end (final)

3. Data inputs (FR-DATA)

- **FR-DATA-01 (M)** Inputs: depth messages (side, price, displayed size, at least the top of book); trades (price, size, aggressor, feed time); instrument metadata (price step, exchange, product code).
- **FR-DATA-02 (M)** If the connection doesn't provide the aggressor side, Hold-Check is unavailable for that instrument. The chip reads "Unavailable on this feed", Settings explains why, and no events are produced. **Classic keeps today's behaviour on every feed.** Signal source (*Eng to confirm*): connector capability metadata. Otherwise, all trades lacking an aggressor for 60 s.
- **FR-DATA-03 (M)** Messages are processed in feed order. Identical message sequences give identical events (NFR-01).
- **FR-DATA-04 (M) Interruption** = connection lost, feed sequence gap, instrument reset, or full-book snapshot, as reported by the platform. Open episodes without an event are discarded. Live events become INTERRUPTED (DATA_GAP) with t_state = the feed time of the last message before the interruption. Messages with t < t_resume + 5 s are ignored. INTERRUPTED events don't block new events at their level.
- **FR-DATA-05 (M) History.** Messages delivered before the platform's real-time-start signal are history (*Eng to confirm* the callback). Events triggered in history carry history=true and none of their transitions alert. On aggregated backfill (e.g. dxFeed) there is no detection on history, and Settings says so.
- **FR-DATA-06 (S)** While Bookmap's automatic data aggregation is active, new events carry aggregated=true, are drawn semi-transparent and are labelled "reduced accuracy". This needs a platform signal (the 8.0 add-on API adds aggregation start/end times; *Eng to confirm*). If none exists, P2.

4. Presets, display and alerts

Presets, sensitivity and sessions (FR-PRE)

- **FR-PRE-01 (M)** Built-in, versioned presets: "ES · Bookmap default v1.0" and "CL · Bookmap default v1.0" (§2).
- **FR-PRE-02 (M)** The preset is chosen from a per-connector mapping of (exchange, product code) for outright futures only: ES on CME, CL on NYMEX. Spreads and options get Classic only.
- **FR-PRE-03 (M)** Sensitivity Low/Medium/High (default Medium), shown in plain language, e.g. "flags ≥ 600 contracts absorbed at one price within 60 s".
- **FR-PRE-04 (M)** Advanced parameters are editable and validated against the allowed range. Out-of-range input is rejected inline with the range shown, and OK is disabled. Editing a parameter creates "Custom (from <preset>)"; sensitivity and direction don't. Reset restores the source preset.
- **FR-PRE-05 (M)** Detection settings are stored in the user's Bookmap account keyed by (exchange, product), so they survive rollover and follow the user across devices. The preset library and display/alert settings live in the same account.
- **FR-PRE-06 (M)** alert_hours per preset. Outside it, markers are still drawn and the throttle isn't used.
- **FR-PRE-07 (M) Partner presets are bundled** in the build ("<Partner> ES" and "<Partner> CL"), pre-loaded at launch with starting values tuned per instrument (§2, Appendix B) plus the partner's alert hours, so cohort students get a working setup with zero configuration from day one. Values are refined to the partner's calibrated numbers by v1.0 without renaming the preset or changing a student's saved settings (Product Requirements §5). They show "by

«Partner» ✓ Published by a verified educator account" wherever the preset name appears. The badge tooltip reads: "Bookmap verifies who published this preset, not its trading results."

- **FR-PRE-08 (S)** On MES/MCL charts, detection runs by default on the parent ES/CL book (same price scale) and draws on the micro chart; the chip reads "reading ESZ6 order flow". Fallback: own-book presets MES 500 / MCL 100 (*calibrate*).
- **FR-PRE-09 (S)** Auto threshold for other instruments reuses today's SD method. The threshold is recomputed at each SD-interval boundary (feed time); no event fires before the first value ("Warming up"). The value in force is snapshotted.
- **FR-PRE-10 (S)** News-mute windows per preset. CL default: EIA report, Wednesday 10:28–10:33 ET (Thursday in holiday weeks). ES: 08:30 and 10:00 ET $\pm$ 2 min, off by default. Inside a window, markers are flagged "news", no alerts fire, and the throttle isn't used.

Heatmap visualisation (FR-VIS)

- **FR-VIS-01 (M)** Zone marker: the band from t_{start} to t_{last} (live: to now). A 15% fill is drawn under the volume dots so the evidence stays visible; a 1.5 px outline goes on top. Minimum height 10 px. Eng to confirm whether the web heatmap chart supports a custom marker/overlay layer at all (§5); if not, MVP ships as a Market Pulse Web widget (list view: price, time, outcome) instead of an on-chart mark, and the heatmap overlay becomes a fast-follow.
- **FR-VIS-02 (M)** Chip at $t_{trigger}$, joined by a dotted leader: ▲ below the zone for buyers absorbing, ▼ above for sellers absorbing. It shows V_{total} (abbreviated from 10 000, e.g. 12.4k), "T2+" for retests, and the state glyph (↗ ✓ × − !).
- **FR-VIS-03 (M)** Styles: Absorbing = side-colour outline with a pulse $\leq$ 6 s; Held = filled chip; Broken = grey dashed; Expired = grey; Interrupted = amber.
- **FR-VIS-04 (M)** Colours: buyers #19E3B1, sellers #FF5FA2, clear of the heatmap palette and volume dots. Direction is also shown by arrow and position.
- **FR-VIS-05 (M)** The tooltip shows the fields in FR-DET-14, the preset with its author, the data mode and any flags. (*Eng to confirm* marker hover; fallback in §5.)
- **FR-VIS-06 (S)** Building: dashed band with "building NN%" or "confirming...", shown only for episodes that could still trigger (conditions 2–5 pass). No alert.
- **FR-VIS-07 (S)** Price-axis tag for live events.
- **FR-VIS-08 (M)** Markers outside the viewport are culled. When zoomed out, chips within 12 px merge into "xN". At most 500 markers per instrument.

Alerts (FR-ALR)

- **FR-ALR-01 (M)** Triggers: Absorbing (default on); Held and Broken (opt-in). Reacting (S, opt-in). Presented in plain language as two separate choices — "tell me the moment heavy absorption starts" (Absorbing) and "also tell me once the level has actually held or broken" (Held/Broken) — so an early, unresolved signal is never shown as if it were already a confirmed reversal.
- **FR-ALR-02 (M) [D]** Absorption declares 5 alert types per instrument to the central alert system: Absorbing-buyers, Absorbing-sellers, Reacting, Held, Broken. They appear in *File* › *Alerts* › *Configure alerts*, the FR-ALR-01 toggles and that window show the same setting, they're written to Alerts History (not in Replay), and they respect Mute All for sound and voice. Template variables:
 - existing: \$INSTR, \$SIDE, \$PRICE;
 - \$VOLUME: $V_{trigger}$ for Absorbing, V_{total} otherwise;
 - \$DURATION: $t_{trigger} - t_{start}$;
 - new: \$STATE, \$MOVE, \$TEST.
- **FR-ALR-02W (M) [W]** On web, the same five alert types are configured in a Settings › Alerts panel (there is no "File › Alerts"): in-app toast (FR-ALR-03), sound after a first click (browser autoplay policy), and a browser notification once granted (FR-ALR-10). A single "Mute all" covers every open instrument, matching Market Pulse Web's existing pattern. Desktop keeps FR-ALR-02 unchanged as a Phase 2 parity target.
 - existing: \$INSTR, \$SIDE, \$PRICE;
 - \$VOLUME: $V_{trigger}$ for Absorbing, V_{total} otherwise;
 - \$DURATION: $t_{trigger} - t_{start}$;
 - new: \$STATE, \$MOVE, \$TEST.
- **FR-ALR-03 (S)** An in-chart toast (top-left, never over the recent 15% of the time axis) with at most 2 visible and "xN more". For Absorption it replaces the central pop-up, so there is no double pop-up.

- **FR-ALR-04 (M)** Alerts cover every open instrument with Absorption on. Each alert names the instrument, and tabs with unseen alerts show a badge.
- **FR-ALR-05 (M)** Throttle: at most N Absorbing alerts per instrument in feed time ($t - 60\text{ s}, t]$. Default $N = 5$; options 1, 2, 3, 5, 10. Suppressed events are still drawn and logged. Outcome alerts are not throttled (at most one per event). Alert fields are excluded from determinism comparisons.
- **FR-ALR-06 (M)** Distinct default sounds for buyers and sellers, with a Test button.
- **FR-ALR-07 (M)** Copy describes and never directs. Example: "▲ Passive buyers absorbing · ESZ6 6,742.50: 612 sold into the bid in 25 s. Level holding." Never "buy", "sell", "reversal" or "high probability" (Product Requirements §8).
- **FR-ALR-08 (M)** Latency and history rules are in NFR-02. In Replay, alerts are sound and toast only, labelled "Replay".
- **FR-ALR-09 (P2) [D]** Not a web MVP blocker (the defects are in today's desktop Classic indicator, which this feature doesn't touch at launch). Reproduce on the current Production and Beta builds ("Add alert" does nothing for Absorption; alert settings disappear from Configure alerts) and fix as part of the desktop-parity port.
- **FR-ALR-10 (M) [W]** Browser notification while the tab is unfocused or backgrounded (not fully closed), via the Notification API, after a one-time permission grant; clicking it focuses the tab and navigates to the event. If the tab is fully closed, nothing fires until Web Push ships (FR-ALR-11, P2). Native OS notification on desktop is a Phase 2 parity item (Eng to confirm Java support).
- **FR-ALR-11 (P2)** Push, one-click Telegram (Bookmap bot) and e-mail. Needs server-side detection and data-licensing sign-off.

Settings and onboarding (FR-UX)

- **FR-UX-01 (M)** Entry points: a new "Absorption" algorithm in Market Pulse Web's "+ Add widget" flow, or a new "Absorption" section in web Settings if Eng finds that the better fit (§5). S: a chart chip showing the mode, preset and author, and sensitivity. Desktop keeps Settings › Configure add-ons › Absorption in Phase 2.
- **FR-UX-02 (M)** First enable shows a one-screen explainer: what it detects and what it doesn't, the marker legend, the preset choice (Bookmap defaults, or "«Partner» ES/CL" — pre-selected for anyone opening Bookmap via the partner's cohort link), "turn on for all open instruments" (on by default), and the disclaimer. "Customize first" opens Settings with the indicator pre-enabled.
- **FR-UX-03 (M)** Existing Classic users stay on Classic, with a one-time banner offering Hold-Check.
- **FR-UX-04 (M)** Not entitled (plan without Absorption): presets can still be previewed and saved. The chip reads "Absorption requires Global+ · Start trial" and no detection runs. The saved preset applies once the user is entitled.
- **FR-UX-05 (M)** Sections: Detection, Display, Alerts, Presets & sharing, About. UI strings are externalised.
- **FR-UX-06 (S)** "What would have fired": a dashed, non-alerting preview layer with a count, computed from platform history (Eng to confirm). It never rewrites live events, the log, feedback or alerts.

5. Compatibility and platform

Classic and Broadcasting API compatibility (FR-CMP)

- **FR-CMP-01 (M)** Classic mode is today's logic, unchanged.
- **FR-CMP-02 (M)** Classic keeps computing and publishing on its existing Broadcasting provider whenever a consumer (e.g. Market Pulse "Absorption Pressure") is subscribed, whatever mode the user displays. NFR-03 budgets include both computations.
- **FR-CMP-03 (S)** Hold-Check events go out on a new provider, `absorption.holdcheck/1` (Appendix A schema), one message per transition, so vendors can build on them.
- **FR-CMP-04 (M) [W]** Hold-Check ships as a new algorithm/widget inside Market Pulse Web ("Absorption"), alongside its existing Volume Pressure, Volume Pressure Imbalance and Price Change — reusing the add-widget flow, per-widget sound thresholds, Mute all and workspaces rather than new UI chrome (§5, Eng to confirm). It also draws on the main heatmap chart wherever FR-VIS-01's overlay capability allows.

Replay and history (FR-RPL)

- **FR-RPL-01 (M)** The engine is deterministic: replaying the same recorded message sequence offline (the §7 fixture harness) gives identical results to live processing (NFR-01). Web has no client-facing Replay at launch, so this is verified via the offline harness, not a Replay UI; desktop's Replay gets Hold-Check in Phase 2.

- **FR-RPL-02 (M)** Calibration labelling (Product Requirements §5) uses desktop’s existing Replay and Notes features against recorded BookmapData sessions — labelling is an internal accuracy exercise and doesn’t require the shipped web product to have Replay. Nothing new is built for it.

Web vs desktop

	Web (MVP)	Desktop (Phase 2 · parity)
Detection	Server-side, one engine per instrument (Market Pulse Web’s algorithm-hosting model) — every connected client sees the same events; alerts don’t depend on a desktop app being open	Same server-side engine, surfaced in the existing Absorption indicator (add-on framework) for parity with current Global+ users
Data	BookmapData CME Group; the web client lists CL (NYMEX) — confirm entitlement (week 0, high priority)	Any connection with aggressor-side trades, plus BookmapData CME or Rithmic for MBO
Where	A new “Absorption” algorithm/widget in Market Pulse Web, or a new Settings section if Eng finds that the better fit (§5)	Settings › Configure add-ons, plus the chip (today’s Absorption indicator)
Configuration	Presets and sensitivity first; Advanced under “More”	Presets, sensitivity, full Advanced panel
Alerts	In-app toast, sound (after one click, per browser policy), browser notification (permission) while the tab is open; Web Push for a closed tab is Phase 2	Central alert system; native OS notification; Telegram (Phase 2 parity)
Today	Heatmap, volume dots, Time & Sales and Market Pulse Web’s 3 algorithms (Volume Pressure, Volume Pressure Imbalance, Price Change) with basic sound alerts — no absorption-type indicator yet	Absorption (Classic), central alerts, Replay — unaffected; gets Hold-Check in Phase 2

Parity rule: the same recorded message sequence gives the same events on every engine. Different connections for one instrument are not expected to match.

Platform capabilities required (confirm in week 0)

Capability	Needed for	Known?	Fallback
Market Pulse Web’s algorithm-hosting API accepts a new custom algorithm, not just its 3 built-in ones	FR-CMP-04, the whole architecture	Eng to confirm — the single biggest week-0 unknown	Ship as a standalone “Absorption” section in web Settings, calling the same server engine directly
Custom marker/overlay on the main web heatmap chart, pushed from a server event stream	FR-VIS-01	Eng to confirm	MVP ships as a Market Pulse Web widget (list view) without an on-chart mark; heatmap overlay is a fast-follow
Browser Notification API already wired into web.bookmap.com or Market Pulse Web	FR-ALR-02W, FR-ALR-10	Likely — Market Pulse Web already uses browser sound alerts	In-app toast only until confirmed
Websocket channel for pushing detection events to connected web clients	FR-DET-01, FR-ALR	Documented (Market Pulse Web’s own websocket API)	—
CL (NYMEX) data	FR-PRE-02, the whole CL	Eng / Data to confirm	Ship ES only at launch; add

entitlement on web.bookmap.com specifically	scope	(week 0, high priority)	CL once confirmed
Account-based settings storage for a new algorithm's presets	FR-PRE-05	Eng to confirm	Store client-side (localStorage) until account storage is wired up
Whether Global+ plan-gating applies the same way on web as on desktop	FR-UX-04	Business / Eng to confirm	Assume yes; revisit if web pricing tiers differ
Feed-time heartbeat / deadline firing on a server process (no desktop heartbeat callback to reuse)	FR-DET-02	Eng to confirm	Fire deadlines on the next message only (slightly coarser timing)
Aggressor capability metadata; non-regular trade flags	FR-DATA-02, FR-DET-03	Eng to confirm	60 s heuristic; include all trades

6. Non-functional requirements

- **NFR-01 Determinism.** Identical message and setting-change sequences give identical events (id, state, reason, times, volumes), across runs, replay speeds and platforms. Alert fields are excluded.
- **NFR-02 Latency.** Measured end-to-end for the server-side path (feed message → engine → websocket push → client render): event visible client-side ≤ 250 ms p99; toast or sound start ≤ 750 ms p95 (the network hop makes this a looser budget than a local desktop engine). Desktop parity (Phase 2) keeps the original in-process budget: event emitted ≤ 50 ms p99. Measured over the AC-35 replays at 1x.
- **NFR-03 Performance.** Reference: the server engine, not a desktop client — one engine instance per instrument serves every connected browser (a real efficiency: a whole cohort watching ESZ6 costs one engine, not N). With 8 instruments running and 200 concurrent viewers, replaying ESZ6 09:30–09:45 ET and CLX6 10:25–10:40 ET on an EIA Wednesday:
 - server CPU/memory scale sub-linearly with viewer count (one engine per instrument, not per viewer);
 - browser tab: median FPS drop ≤ 5% with the widget/overlay active;
 - browser tab heap at most +20 MB per instrument after 6 h; server heap growth is bounded per engine instance;
 - at 5 000 events, the oldest final events are evicted.
- **NFR-04 Scale.** The server hosts one engine per actively-subscribed instrument system-wide, not per user. The cap applies to the browser client: at most 8 instruments open in one tab; a 9th is blocked with a message.
- **NFR-05 Resilience.** An engine error never crashes the client. Events become INTERRUPTED (ENGINE_ERROR); the server-side engine instance restarts within 5 s and clients reconnect automatically. After 3 errors in a session it stays paused, with a message shown to connected clients.
- **NFR-06 Accessibility.** Markers work without colour, alerts have text, and sound is optional.
- **NFR-07 Safety.** Presets are data only: validated, and rendered as text.
- **NFR-08 Data licensing.** No new market data leaves the platform in the MVP. Today's Telegram behaviour in Classic is unchanged.

7. Acceptance criteria

Golden scenarios (engine)

These are deterministic fixtures. QA owns them as JSONL input files (`{t_ns, type, side, price_ticks, size, aggressor}`) plus expected-event files, due by the end of week 1. The prototype's Spec tests are the executable reference.

Parameters: the ES preset, with p = the bid level (e.g. 6,742.50) and q = the ask level; the CL preset for AC-23. Times are in seconds. Unless stated otherwise, displayed size = 100.

ID	Given	Expected
AC-01 Held	34 SELL×18 at p every 0.75 s, t = 1.00–25.75.	Threshold at trade 34 (V = 612). ABSORBING

	Displayed bid S0 = 180, then refilling 40–220. BUYs at p+1 (40), p+2 (42), p+4 (44), p+6 (48), p+8 (52)	at 28.75 ; REACTING at 44; HELD at 52. t_last 25.75, t_end 45.75 (GAP). One alert
AC-02 Broken	13 BUY×50 at q every 2.4 s (ask 200). BUY at q+1 at 41, BUY at q+2 at 42	V = 600 on trade 12 (t = 26.4). ABSORBING at 29.4. BROKEN (TRADED_THROUGH) at 42. With Broken alerts on, exactly 2 alerts
AC-03 Confirm window	AC-02's first 12 trades, then BUY at q+2 at 27.9	No event, no alert
AC-04 Poke reclaimed	AC-02 through ABSORBING; BUY q+1 at 41, BUY q at 41.8	Still ABSORBING, pokes = 1
AC-05 Poke not reclaimed	AC-02 through ABSORBING; BUY q+1 at 41; nothing back at ≤ q	BROKEN (NOT_RECLAIMED) at 44
AC-06 Poke before trigger	25 SELL×20 at p (t = 0–24); SELL p–1 at 24.5; SELL×10 at p at 25.2; 5 SELL×20 at p, t = 26–30	Event with pokes = 1, V_trigger 610, ABSORBING at 33
AC-07 Unreclaimed poke before trigger	As AC-06 without the reclaim; then 10 SELL×20 at p, t = 30–39	No event (the second episode starts from 0)
AC-08 Too little	59 SELL×10 at p over 58 s	No event. (S) Building shown from V ≥ 300
AC-09 Slow grind	70 SELL×10 at p spread evenly over 150 s	No event
AC-10 Gap	20 SELL×20 at t = 0–9.5; nothing until 34.5; 15 SELL×20 at t = 34.5–41.5	No event; second episode V = 300 (the combined 700 would trigger without the gap rule)
AC-11 Being eaten	30 SELL×20 at p (t = 0–29); displayed size falls 800 → 220	No event (ratio 0.75; 27% left)
AC-11b Wall intact	As AC-11 with the displayed size falling 2 000 → 1 420	Event (71% left)
AC-12 Spoof	Displayed 900; 10 SELL×10; then SELL at p–2 at 12	No event
AC-13 One live event	AC-01 to ABSORBING. Price stays at p+1...p+2. A new SELL episode at p (t = 100–124.75) reaches 600	Still 1 event (ABSORBING)
AC-14 Retest	AC-01 (HELD). Then 34 SELL×18 at p from t = 90	Second event at 117.75, test_n = 2
AC-15 After Broken	AC-01 to ABSORBING. SELL at p–2 at 35 (BROKEN). 34 SELL×18 at p from 60	Second event, test_n = 1
AC-16 De-dup	AC-01 to ABSORBING. 30 SELL×21 at p+1, t = 30–44.5	No new event
AC-17 Throttle	5 levels each reaching 600 in 10 s, at t = 0, 11, 21, 31, 70 (bid p, ask p+10, bid p+5, ask p+15, bid p+20)	5 events. Alerts for 1–3; 4th suppressed; 5th alerts (window rolled)
AC-18 Expiry	AC-01 to ABSORBING. BUYs at p+1...p+3 every 10 s to t = 340	EXPIRED (TIMEOUT) at 328.75
AC-19 High	AC-08 data at High (threshold 360)	V_trigger 360; ABSORBING at 38
AC-20 Direction	AC-01 with "Sellers absorbing" only	No bid event
AC-21 Min visible	min_visible 150. AC-01 timings with size 100, reaching 160 at trade 10 → event. Never above 120 → none	1 event; 0 events
AC-22 Unknown aggressor	AC-01 with trades 10 and 20 UNKNOWN → none (576). An UNKNOWN trade at p–2 after ABSORBING → BROKEN	As stated
AC-23 CL band	CL preset: 38 SELL×10 cycling 71.42 / 71.41 / 71.40 over 49 s	One event, band 71.40–71.42
AC-24 Interruption	Open episode + gap → discarded. ABSORBING + gap → INTERRUPTED (DATA_GAP)	As stated
AC-25 Freeze	AC-01 to ABSORBING; BUY p+8 at 30 (HELD); SELL×20 at p each second, t = 33–45	Stays HELD at 30; V_total frozen at 612; no new event
AC-26 Determinism	Scenarios run with no heartbeat and with heartbeats of 0.1 s and 1 s; Replay 1× and 10×	Identical event lists
AC-27 Boundaries	33×18 + 5 = 599 → none; + 6 = 600 → event. A lot exactly 60 s old is excluded. A trade exactly at t_last + 20 continues the episode	As stated

UX and integration acceptance

- **AC-30 Alert system.** Absorption alert types appear in the web Settings › Alerts panel, are logged in the Absorption log, and obey Mute all. (Desktop parity, Phase 2: the same types also appear in Configure alerts, write to Alerts History, and the FR-ALR-09 defects don't reproduce.)
- **AC-31 First run.** Starting from a Global+ account with an ES chart and Absorption never enabled: in ≤ 60 s, ≤ 3 clicks and no typing, Hold-Check is on with Absorbing alerts, confirmed by the Test sound. 5 participants, ≥ 4 succeed.
- **AC-32 Attribution.** The partner preset shows "by <Partner> ✓ ..." in settings, the chip, the tooltip and the log. A shared preset naming the partner shows "(unverified)". A tampered code, out-of-range values, an HTML name and an instrument mismatch are each rejected or rendered as specified.
- **AC-33 Classic compatibility.** With the new Absorption algorithm live in Market Pulse Web, its existing Volume Pressure, Volume Pressure Imbalance and Price Change algorithms produce identical output to production over 3 recordings. (Desktop parity, Phase 2: Market Pulse Absorption Pressure output and Classic markers/alerts stay identical to production.)
- **AC-34 Not entitled and persistence.** A non-Global+ user sees the trial state (FR-UX-04). Custom settings survive the ESZ6 → ESH7 rollover.
- **AC-35 Performance.** The NFR-02/03 measurements pass on the named recordings.

Traceability: FR-DET → AC-01...27. FR-ALR → AC-17, AC-30. FR-SHR/PRE → AC-32, AC-34. FR-CMP → AC-33. FR-UX → AC-31, AC-34. NFR → AC-26, AC-35. FR-VIS and FR-LOG values (sizes, counts, timings) are checked as written, via a UI checklist.

Appendix A: Event schema

```
{ "id": "ESZ6|BID|26970|1790343667750000000", "instrument": "ESZ6", "exchange": "CME",
  "side": "BID", "band": { "near": 6742.50, "far": 6742.50, "ticks": 1 }, "mode": "HOLDCHECK",
  "state": "HELD", "state_reason": "MOVED_AWAY", "test_n": 1, "pokes": 1,
  "t_start": "2026-09-25T13:40:41.000Z", "t_trigger": "2026-09-25T13:41:07.750Z",
  "t_last": "2026-09-25T13:41:04.750Z", "t_end": "2026-09-25T13:41:24.750Z", "end_reason": "GAP",
  "t_state": "2026-09-25T13:41:31.000Z", "v_trigger": 612, "v_total": 612, "s0": 180, "s_max": 220,
  "replenishment_ratio": 2.7818, "max_move_away_ticks": 8,
  "flags": { "history": false, "aggregated": false, "aggressor_inferred": false },
  "alerts": [ { "kind": "ABSORBING", "t": "2026-09-25T13:41:07.750Z", "suppressed": false } ],
  "data_mode": "MBP", "engine_version": "1.0.0",
  "preset": { "id": "es-default", "version": "1.0", "author": "Bookmap", "verified": true, "sensitivity": "Medium",
    "effective": { "min_volume": 600, "window_s": 60, "max_gap_s": 20, "zone_ticks": 1, "confirm_s": 3,
      "min_ratio": 1.0, "min_remaining": 0.5, "break_ticks": 2, "reclaim_s": 3, "min_visible":
0,
      "react_ticks": 4, "held_ticks": 8, "outcome_timeout_s": 300 } } }
```

Appendix B: Presets (all *calibrate*)

Preset	min_volume	band	confirm	break / reclaim	react / held	alert hours (ET)
ES · Bookmap default v1.0	600 / 60 s	1	3 s	2 ticks / 3 s	4 / 8	09:30–16:00
CL · Bookmap default v1.0	300 / 60 s	3	5 s	3 ticks / 3 s	5 / 15	09:00–14:30 (EIA mute, S)
MES / MCL (fallback, FR-PRE-08)	500 / 100	1 / 3	as ES / CL	as ES / CL	as ES / CL	as ES / CL
<Partner> ES (bundled, launch-ready)	650 / 60 s	1	3 s	2 ticks / 3 s	4 / 8	09:30–11:30
<Partner> CL (bundled, launch-ready)	330 / 60 s	3	5 s	3 ticks / 3 s	5 / 15	09:00–11:00 (EIA mute)

Where the starting values come from: ES 600 in 60 s is about 2–3× typical peak-hour displayed depth per level (often 250+ contracts). That is in line with the "≈ 3× average depth" rule of thumb other order-flow tools use for large-quantity alerts. CL is scaled to its thinner book and summed across 3 ticks. These are guesses made without market data, for Product Requirements §5 to replace.